M. Ryleigh Davis

Postdoctoral Researcher in Planetary Science

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Education

Ph.D. **California Institute of Technology**, Planetary Science — Sept 2020 to Apr 2026

- **Thesis:** Spectroscopic Characterization of Icy Moon Surfaces: Compositions, Origins, and Implications from Jupiter to Neptune **Advisor: Mike Brown**

M.Sc. **California Institute of Technology**, Planetary Science — May 2022

M.Sc. **Northern Arizona University**, Applied Physics — Aug 2018 to May 2020

- **Thesis:** Low Energy Electron Sputtering of Water Ice **Advisor: Mark Loeffler**

B.Sc. **University of Arizona**, Astronomy & Physics, minor in Mathematics — Aug 2013 to May 2017

Work & Research Positions

Postdoctoral Researcher, UC San Diego - Department of Astronomy & Astrophysics — CA, USA, May 2026 to present

- Lead multi-wavelength (UV–mid-IR) observational programs using HST, JWST, and ground-based facilities to characterize the surface composition of icy satellites across the outer solar system. **Faculty host: Samantha Trumbo**

Senior Associate Consultant, Clarity Insights: Data Analytics Consulting — IL, USA, July 2017 to Aug 2018, 1 year 1 month

- Lead the team developing a platform for distributed computations on the cloud (via pySpark and AWS) for a major financial institution and worked with the Data Science team to implement machine learning and predictive algorithm tools for data analytics.

Telescope Operator, University of Arizona - Steward Observatory — AZ, USA, Jan 2014 to May 2017, 3 years 5 months

- Operated telescopes for research data acquisition and trained/certified new observers (Bok 2.2m/B&C Spec, Bok 2.2m/90Prime, Kuiper 1.55m/Mont4k, Ray White 21").
- Assisted Mountain Operations with Instrument changes, system testing, cleaning of mirrors and filters and other telescope maintenance projects as a summer intern.

NASA Space Grant, University of Arizona, Research Intern — AZ, USA, Aug 2015 to May 2016, 1 year

- Worked with Dr. Caitlin Griffith to observe and reduce Kuiper 1.55m telescope data to examine and characterize the atmospheres of hot Jupiter exoplanets

Selected Awards

- **Academic Excellence in Research Award**, Caltech Division of Geological and Planetary Sciences (2026)
- **Graduate Fellow**, Caltech Center for Comparative Planetary Evolution (3CPE) (2021-2026)
- **Alexander F. H. Goetz Fellowship**, Caltech (2020-2021)
- **Grad College/GSG Academic Scholarship Award**, Northern Arizona University (2019)
- **Undergraduate Research Achievement**, University of Arizona Department of Astronomy (2017)
- **Arizona Excellence Award**, University of Arizona (2013-2017)

Invited Seminars & Colloquia

- **UC Santa Cruz**, PLUNCH Seminar, May 18, 2026
- **Carnegie EPL**, Astronomy Seminar, Mar. 27, 2026
- **University of Arizona**, Origins Seminar Series, Mar. 9, 2026
- **UC San Diego**, Astronomy & Astrophysics Colloquium, Mar. 11, 2026
- **UCLA**, Astronomy Lunch Seminar, Dec. 9, 2025
- **Uranian Rings & Small Satellites Working Group**, Science Talk, Dec. 5, 2025

- **JPL**, Icy Worlds Collaboration and Exchange (ICE) Seminar, Apr. 10, 2025
- **Europa Clipper Team**, Composition Working Group Science Talk, Jan. 23, 2024
- **Caltech**, DIX Planetary Science Seminar, Jun. 7, 2022; May 9, 2023; Jun. 4, 2024; Apr. 15, 2025

Publications

First-Author

- **Davis, M.R.**, Belyakov, M., Wong, I., Milby, Z., and Brown, M.E (2026). "Neptune's Inner Moons and Rings Are Exposed Icy Body Interiors". Science Advances (in press).
- **Davis, M.R.** (2026). "Spectroscopic Characterization of Icy Moon Surfaces: Compositions, Origins, and Implications from Jupiter to Neptune". (Ph.D. Thesis, California Institute of Technology). DOI: 10.7907/605z-1m76.
- **Davis, M.R.**, Trumbo, S.K., Brown, M.E, and Belyakov, M. (2025). "Spectroscopic Mapping of Callisto with HST/STIS and Implications for its Surface Composition". The Planetary Science Journal, 6, 161.
- **Davis, M.R.** and Brown, M.E. (2024). "Pwyll and Manannán Craters as a Laboratory for Constraining Irradiation Timescales on Europa". The Planetary Science Journal, 5, 107.
- **Davis, M.R.**, Brown, M.E., & Trumbo, S.K. (2023). "The Spatial Distribution of the Unidentified 2.07 µm Absorption Feature on Europa and Implications for its Origin". The Planetary Science Journal, 4(8), 148.
- **Davis, M.R.**, Meier, R.M., Cooper, J.F., and Loeffler, M.J. (2021). "The contribution of electrons to the sputter-produced O2 exosphere on Europa". The Astrophysical Journal Letters, 908(2), L53.
- **Fitzpatrick (Davis), M.R.** (2020). "Low Energy Electron Sputtering of Water Ice" (M.Sc. Thesis, Northern Arizona University).

Co-Authored

- Carmack, R.A., **Davis, M.R.**, and Loeffler, M. (2026). "Enhanced surface erosion of icy moons due to ice structure". (submitted to PSJ).
- Belyakov, M., Wong, I., Lisse, C.M., **Davis, M.R.**, Bolin, B.T., Martin, A., Pontoppidan, K.M., Blake, G.A., Chen, C., and Brown, M.E. (2026). "The Dust Mineralogy of Interstellar Comet 3I/ATLAS from JWST/MIRI Observations". *The Astrophysical Journal Letters* (in review).
- Belyakov, M., **Davis, M.R.**, Wong, I., Batygin, K., and Brown, M.E. (2026). "Nereid as a Regular Satellite of Neptune". *Science Advances*, 12(21), eaeb1429.
- Belyakov, M., Wong, I., Bolin, B.T., **Davis, M.R.**, Bromley, S.J., Lisse, C.M., and Brown, M.E. (2026). "The Volatile Inventory of 3I/ATLAS as seen with JWST/MIRI". *The Astrophysical Journal Letters*, 1001(1), L11.
- Brown, M.E., Belyakov, M., Chandra, S., **Davis, M.R.**, McDowell, M., Pandya, A., Trinh, K., and Trumbo, S.K. (2026). "Deuterated water and the formation of the satellites of Uranus". *Proceedings of the National Academy of Sciences*, 123(27), e2519276123.
- Brown, M.E, Trumbo, S.K., **Davis, M.R.**, and Chandra, S. (2025). "Deuterated water on the satellites of Saturn". The Planetary Science Journal, 6, 229.
- Brown, M.E, Trumbo, S.K., Belyakov, M., **Davis, M.R.**, and Pandya, A. (2025). "A JWST study of CO_2 on the satellites of Saturn". The Planetary Science Journal, 6, 180.
- Belyakov, M. et al. (inc. **Davis, M.R.**) (2025). "Palomar and Apache Point Spectrophotometry of Interstellar Comet 3I/ATLAS", Research Notes of the AAS, 9, 194.
- Belyakov, M., **Davis, M.R.**, Milby, Z., Wong, I., and Brown, M.E. (2024). "JWST Spectrophotometry of the Small Satellites of Uranus and Neptune". The Planetary Science Journal, 5(5), 119.
- Kunimoto, M., Vanderburg, A., Huang, C.X., **Davis, M.R.**, et al. (2023). "TOI-4010: A System of Three Large Short-period Planets with a Massive Long-period Companion". The Astronomical Journal, 166(1), 7.
- Trumbo, S.K., **Davis, M.R.**, Cassese, B., and Brown, M. E. (2022). "Spectroscopic Mapping of Io's Surface with HST/STIS: SO2 Frost, Sulfur Allotropes, and Large-scale Compositional Patterns", The Planetary Science Journal, 3(12), 272.
- Zellem, R.T. et al. (inc. **Fitzpatrick (Davis), M.R.**)(2015). "XO-2b: A hot jupiter with a variable host star that potentially affects its measured transit depth", The Astrophysical Journal, 810(1), 11.

Grants and Observing Programs

PI and Science PI Programs:

- Unlocking Europa's Ocean Chemistry: Disentangling Native Salts from Radiolytic Cycles with JWST/MIRI (PI); JWST/MIRI Cycle 5 GO (2026-2027); 4.6 hrs
- Tracing the Origin of Hydrated Minerals on the Small Satellites of Uranus and Neptune (PI); Palomar/NGPS (2025B); 5 nights
- Constraining the Composition and Thermal Histories of Silicate Minerals on Callisto (PI); JWST/MIRI Cycle 3 GO (2024-2025); 15.96 hrs
- Reconstructing the Histories of the Ice Giant Systems through Small Satellite Observations (Co-PI); JWST/NIRSpec Cycle 3 GO (2024-2025); 26.71 hrs
- Photometric U and B Band Observations of Bright Transiting Exoplanets to Constrain their Atmospheric Compositions (Undergraduate PI); Kuiper/Mont4K & Bok/90Prime (2014-2017); 37 nights

Co-I Programs:

- MIRI MRS Observations of the Third Interstellar Object (PI: Belyakov); JWST/MIRI Cycle 4 DDT (2025); 9.2 hrs
- The Aftermath of a Transient Deposition Event on Europa (PI: Trumbo); JWST/NIRSpec Cycle 3 DDT (2024); 13.2 hrs
- The Saturnian satellites as a laboratory for CO2 in the outer solar system (PI: Brown); JWST/NIRSpec Cycle 2 GO (2023-2024); 21.2 hrs
- Complete NIRSpec coverage of Europa's surface: CO2, salt hydrates, and the potential for unexpected discovery (PI: Trumbo); JWST/NIRSpec Cycle 2 GO (2023-2024); 21.2 hrs
- Titan's Tropospheric Methane Abundance Across the Seasons (PI: Griffith); NASA Solar System Workings (2019-2022)
- A Global Map of Titan's Tropospheric Methane Abundance Near Northern Solstice (PI: Griffith); Keck II/NIRSPEC (2017-2019); 10 half nights

Teaching Positions

Graduate Teaching Assistant, California Institute of Technology — CA, USA; Oct 2021 to Dec 2024

- Teaching Assistant for Ge 108: Applications of Physics to the Earth Sciences (3 terms)

Graduate Teaching Assistant, Northern Arizona University — AZ, USA; Aug 2018 to May 2020

- Instructor for Introductory Electricity and Magnetism laboratory class (4 semesters).
- Teaching Assistant for Indigenous Astronomy (2 semesters).

Student Educator and Video Producer, Steward Observatory Education Group — AZ, USA; Aug 2015 to May 2017; 1 year 9 months

- Communicated with and educated the public on Astronomy and science related topics via production of YouTube videos on the Active Galactic Videos channel.
- Aided in curriculum development, filming, and video editing for an Introductory Astronomy online course (hosted on Coursera) with Professor Chris Impey.

Special Projects Assistant for Education & Public Outreach, National Optical Astronomy Observatory (NOAO) — AZ, USA; Aug 2014 to Oct 2015; 1 year 2 months

- Developed Astronomy and STEM related education curriculum and outreach activities, coordinated and ran public events, managed the Globe at Night citizen science campaign, and ran teacher training's and workshops on Astronomy and Space Education.
- Co-ran an outreach and education training session at the AAS Winter Meeting.

Engineering & Robotics Instructor, Engineering for Kids Summer Camp — AZ, USA; May to Aug 2014; 3 months

- Taught science, engineering, and robotics summer camp curriculum including Lego Robotics for students aged 3-7 and Engineering Basics and Design for students 8-14.

Science Communication & Outreach

- **Astronomy on Tap San Diego**; Public lecture (August 4, 2026)
- **Caltech Everhart Lecture**; Invited public lecture, "A Planetary Cold Case: Using JWST to Uncover the Catastrophic

History of Neptune's Moons" (April 21, 2026) [recording]

- **University of Arizona College of Science Career Center**; Panelist for Earth, Physical, & Space Science Graduate Student Panel (March 19, 2025)
- **Caltech GPS Buddy Program**; Mentor (2021-2025)
- **Caltech GPS Division Planetary Science Option Ombuds Person** (2021-2022)
- **Caltech Women Mentoring Women**; Mentor (2021-2022)
- **Skype a Scientist**; Scientist Partner (2020-current)
- **Letters to a Pre-Scientist**; Scientist/STEM Professional (2022-2024)
- **Society of Women in Space Exploration (SWISE)**; NAU chapter co-founder & Vice President (2018-2020)
- **Middle School Science Mentor**; Sinagua Middle School (Flagstaff, AZ) (2019-2020)
- **Flagstaff Community Star Party**; Public lecture (2019)
- **University of Arizona Astronomy Club**; President & Outreach Coordinator (2013-2017)

Selected Conference Talks

- **Davis, M.R.**, Belyakov, M., Wong, I., Milby, Z., and Brown, M.E. (Dec 2026) "Neptune's Inner Moons and Rings Are Exposed Icy Body Interiors" *[Invited]*, AGU Fall Meeting (San Francisco, CA)
- **Davis, M.R.**, Trumbo, S.K., Brown, M.E., Camarca, M., de Kleer, K., Thelen, A., and Wong, I. (Oct 2026) "Constraining the Composition of Callisto's Dark Material with JWST/MIRI", DPS Meeting #58 abs. 189 (Spokane, WA)
- **Davis, R.** & Brown, M. (Nov 2025) "From UV to Mid-IR: Illuminating the Nature of Callisto's Dark Material", Jovian Icy Moons' Surface Interactions With Their Environment Workshop (Madrid, Spain)
- **Davis, R.**, Belyakov, M., Brown, M., and Wong, I. (Sept 2025) "JWST Reveals Phyllosilicates on the Small Inner Moons of Neptune", EPSC-DPS2025 #181 (Helsinki, Finland)
- **Davis, M.R.**, Belyakov, M., Wong, I., and Brown, M.E. (Jun 2025) "JWST Spectroscopy of the Small Inner Satellites and Rings of Uranus and Neptune", OPAG Meeting 2025 #6008 (Tucson, AZ)
- **Davis, M.R.** and Brown, M.E. (Jan. 2025) "Exploring the Composition of Callisto's Dark Material with JWST MIRI MRS", JWST Solar System Science Workshop (Meudon, France)
- **Davis, M.R.**, Trumbo, S.K., and Brown, M.E. (Dec 2024) "Spectroscopic Mapping of Callisto's Surface with HST/STIS", AGU Fall Meeting P32A-06 (Washington D.C.)
- **Davis, M.R.** and Brown, M.E. (Dec 2023). "Pwyll and Manannán Craters as a Laboratory for Constraining Irradiation Timescales on Europa", AGU Fall Meeting P43B-04 (San Francisco, CA).
- **Davis, M.R.** and Brown, M.E. (Oct 2023). "Pwyll Crater as a Laboratory for Understanding Irradiation Timescales on Europa", Workshop on the Origins and Habitability of the Galilean Satellites (Marseille, France)
- **Davis, M.R.** and Brown, M.E. (Oct 2022). "A Critical Analysis of the 2.07 µm Absorption Feature on Europa", DPS Meeting #54 (London, Canada).
- **Fitzpatrick (Davis), M.R.** and Griffith, C.A. (Jan 2020). "New Reduction Techniques for Echelle Spectroscopy of Extended Objects: Creating a Global Map of Titan's Tropospheric Methane Humidity", AAS #235 (Honolulu, HI).
- **Fitzpatrick (Davis), M. R.** and Loeffler, M. J. (Dec 2019). "The Sputtering Yield of Electron Irradiated Water Ice and Its Dependence on Temperature", AGU Fall Meeting (San Francisco, CA).
- **Fitzpatrick (Davis), M. R.** and Loeffler, M. J. (Sept 2019). "Composition of the Sputtered Flux in Electron Irradiated Ices", Flagstaff Astronomy Symposium (Flagstaff, AZ).
- **Fitzpatrick (Davis), M.R.**, Martins-Filho, W., Griffith, C.A., Pearson, K., Zellem, R.T., and AzGOE (Oct. 2016). "A Study of the Effects of Underlying Assumptions in the Reduction of Multi-Object Photometry of Transiting Exoplanets", DPS Meeting #48 (Pasadena, CA).
- **Fitzpatrick (Davis), M.R.**, Watson, Z., Zellem, R.T., Pearson, K., Griffith, C.A., and AzGOE (Jan 2015). "High-precision ground-based observations of transiting exoplanets to detect their magnetic fields and undiscovered companions", AAS Meeting #225 (Seattle, WA).